

Objectives

- Uncover problems and develop solutions for measurable phenomena (ie. anthropogenic processes - climate change) and their downstream outcomes (ie. ecosystem biology).
- Develop effective methods to discover, enhance, or develop predictive, preventive, and precision medicine for treating or alleviating chronic diseases.
- Engage in ubiquitous environmental and scientific outreach, inspire current generations, and make measurable impacts toward a resilient, connected, and empathetic world!

Skills

- **Programming Languages**
 - Python.
 - Coursework in Java, C++, R, and Ruby.
 - Minor experience with JS, HTML, VueJS, NodeJS, and CSS.
- **Packages**
 - TensorFlow, Matplotlib, Pandas, Scikit-learn, and Seaborn.
- **Interests/Areas of Research**
 - Machine Learning, Neuroscience, Climate Change, Public Health, and Magnetic Resonance Imaging.

Research Experience

Joint Research Scholar - University of Vermont - Center for Complex Systems and Data Science, Computational Story Lab, and Neurobotics Lab
Aug 2023 - Present

- Analyzed U.S. Census and Department of Health & Human Services data using machine learning to identify statewide chronic disease mortality predictors and explore the relationship between socioeconomic factors and health trends. - Solo Project -
 - Research abstract and 2-page paper accepted to and presented at the 2025 Annual American Public Health Conference Meeting & Expo.
- Studied how body composition and psychosocial factors like self-compassion and social support change across seasons in college students and identified key patterns that could help improve mental health interventions and support.



- Received a Summer Undergraduate Research Fellowship grant to continue research during the 2025 summer.
- Tested machine learning models and utilized Canonical Correlation Analysis to identify digital phenotypes and predict preterm and term Preeclampsia.
 - Research Abstract accepted to and presented at the 2024 46th Annual International Conference of the IEEE Engineering in Medicine & Biology Society.
- Designed and tested potential pathways for expanding undergraduate involvement in graduate-level research projects.
- Brainstormed new avenues for improving data collection processes within MassMutual's data science team to advance rigorous, interpretable, and inclusive science for a global audience.
- Learned how to query and run jobs on UVM's computing cluster (supercomputer).

Paid Student Researcher - Stanford School of Medicine - Center for Interdisciplinary Brain Sciences Research, Dr. Hong Lab
Jun 2021 - Jun 2023

- Developed machine learning models, using SciKit Learn and PyTorch, to identify and compare brain regions that determine biological sex in pre- and post-adolescents' structural magnetic resonance images.
 - Presented research at Stanford School of Medicine's Maternal & Child Health Research Institute 2022 Symposium, Moderated Poster session.
- Manually corrected brain tissue segmentation in MRI images using FreeSurfer, an open-source software for processing and analyzing MRI images.
- Completed certification and operated 3T3 & 3T2 MRI machines at the Richard m. Lucas Center for Imaging.
- Performed functional connectivity analysis on resting-state fMRI images of Mosaic Turner Syndrome participants in the Puberty and Neurodevelopment in Adolescence (PANDA) study.
- Customized a Google Cloud virtual machine for remote data analysis and data acquisition.
- Developed a Bash CLI for Freesurfer integration with Sherlock, Stanford's computing cluster (super-computer).

Conference Presentations

- Enerson Poon, Mikaela Fudolig: "Leveraging eXtreme Gradient Boosting and Multiclass Stratification to Analyze the Impact of Sociodemographic Factors on Statewide Chronic Disease Deaths Across U.S. States" *American Public Health Association*, Washington, D.C, November 2025
- Enerson Poon, Mikaela Fudolig, Johanna Hidalgo, Bryn Loftness, Katheryn Stanton, Nick Cheney, Christopher M. Danforth, Connie Tompkins, Matthew Price: "Body Composition Correlates with Self-Reported Social Support and Self-Compassion Among College Students: Insights From the Lived



Experiences Measured Using Rings Study (LEMURS)", *UVM Student Research Conference*, Burlington, April 2025.

- Enerson Poon, Bryn Loftness, Nick Cheney, Carole McBride, Ira Bernstein "Selection Algorithms Identify Pre-Pregnancy Features Associated with Preeclampsia Onset and Birthweight Percentile", *Institute of Electrical and Electronics Engineers Engineering in Medicine and Biology Conference*, Orlando, Florida, July 2024.
- Enerson Poon, Javad Ansarifard, David S. Hong, "Deep Learning Identifies Sex-Specific Regions in Pre- and Post-adolescent Brains", *Stanford School of Medicine's Maternal & Child Health Research Institute 2022 Symposium*, Stanford, October 2022.

Professional Experience

Machine Learning & Data Science Intern - BETA Technologies - Data Team
May 2024 - Aug 2024

- Developed a machine learning pipeline, using SciKit Learn and Pandas, to tag thermal runaway events within time series pressure data. + Gitlab CI for development, automated testing, and deployment.
- Improved software engineering skills: recyclable code, docstrings, MyPy, and Lint.
- Learned how to operate in a high-performance computing environment.

Outdoor Programs Intern - University of Vermont
Jul 2023 - Aug 2023

- Developed trip plan, gear lists, and interpersonal activities for a 23-person camping trip.
- Honed leadership, planning, outdoor recreation, outdoor education, and project management skills.
- Cleaned up and reorganized over 150 lbs of medical supplies.

Software Engineering Intern - Loft Orbital
Feb 2021 - Aug 2021

- Developed and redesigned a 3D visualization component of Loft's satellite control software: displaying satellites, orbits, and ground stations in real-time — using Vite, Lit-Element, VueJs, CSS, HTML, JS, Docker, and Gitlab CI.
- Used Docker and Gitlab CI for development, automated testing, and deployment.
- Improved leadership, project management, and team communication skills.

Information Technology Summer Intern - The Nueva School
Feb 2020 - Aug 2020

- Performed diagnostic repair (OS Repair), mechanical repair (Screen and SSD replacement), and provisioning on MacBook Pros.

Awards

2026 - UVM - Presidential Leadership Fellowship

- *Working with and learning from UVM President Marlene Tromp, Honors College Dean David Jenemann, and fellow industry leaders to understand and uncover the complexities of societal challenges, identify and inquire into the impact of these challenges on multiple scales, and examine and evaluate evidence-based solutions.*

2025 - UVM - Summer Undergraduate Research Fellowship



- Received a financial fellowship to support the further development of my psychosocial research on the Lived Experiences Measured Using Rings Study (LEMURS) over the 2025 summer.

Personal Projects

PovellestoFilms (2020 - Present) ([A Letter To Our Future Selves Short Film](#))

- Pursued a creative project to examine how to capture life authentically using a camera.

PovellestoDraws (2023-2024)

- Started an art business, broke even, and made 7x initial investment by the end of month 1.

Certifications

May 2023 - SDI Open Water Scuba Diver (SDI) #1398432

May 2023 - Computer Nitrox Diver (SDI) #1398426

May 2023 - Reef Renewal Diver (PADI) #23050T8415

Education

2023 - Present - University of Vermont (*expected 2027*)

- Neurobotics & Artificial Intelligence (B.A), Minor in Neuroscience
 - Intro to Machine Learning, Genetics, Molecular Bio, Psych Research Methods w/lab, Exploring Neuroscience w/lab, Learning, Cognition & Behavior

2018 - 2023 - The Nueva School (*HS Diploma - 2023*)

- Advanced Machine Learning, Computer Vision, Software Engineering, Algorithms, Computer Internals, Entrepreneurship, and Data Science.